Voltage dividers & sensors

Higher · Electronics & control · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate V2.

- V2 = Vs × R2 ÷ (R1 + R2) = 6 × 15 ÷ 25
- V2 = 3.6 V

3 Quick check

- a) 6 V
- b) 24 kΩ
- c) The larger share of voltage
- d) The sensor's resistance changing

4 Key terms

- 1 – A
- 2 – F
- 3 – E
- 4 – D
- 5 – B
- 6 – C

5 Calculate Vout across the thermistor at 20 °C.

- Vout = 5 × 2.2 ÷ (4.7 + 2.2)
- Vout = 1.6 V

6 Calculate the value of R1.

- V1 = 6 − 1.5 = 4.5 V
- R1 = R2 × V1 ÷ V2 = 1200 × 4.5 ÷ 1.5
- R1 = 3600 Ω (3.6 kΩ)

7 Calculate the voltage between the two midpoints.

- VA = 12 × 10 ÷ 20 = 6 V
- VB = 12 × 12 ÷ 20 = 7.2 V
- Difference = 1.2 V